

Lab 08 Answers Report

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1 Question 1

Define a Prolog knowledge base for the Unix directory structure. Refer to the diagram on the previous slide. Use the predicate `location(Item, Folder)` to represent that an *Item* (which can be a file or another folder) is located directly inside a *Folder*. For the top-level directory `/`, refer to it as *root*.

1.1 Implementation

```
1 % Directory structure
2 location(bin, root).
3 location(etc, root).
4 location(sbin, root).
5 location(usr, root).
6 location(var, root).
7 location(home, root).
8
9 % /bin dir
10 location(bash, bin).
11 location(cat, bin).
12 location(ls, bin).
13
14 % /etc dir
15 location(crontab, etc).
16 location(cups, etc).
17 location(fonts, etc).
18 location(fstab, etc).
19 location('host.conf', etc).
20
21 % /usr dir
22 location(bin, usr).
23 location(include, usr).
24 location(lib, usr).
25 location(local, usr).
26
27 % /usr/local dir
28 location(bin, local).
29 location(lib, local).
```

Listing 1: UNIX Directory Knowledge Base

2 Question 2

Query the knowledge base to find the parent folder (the direct location) of the `fstab` file. Ensure that you show the exact Prolog query and the results received.

Result:

```
?- ['unix.pl'].
true.

?- location(What, usr).
What = bin ;
What = include ;
What = lib ;
What = local.

?- location(What, local).
What = bin ;
What = lib ;
What = man ;
What = share.

?- location(local, Grandparent).
Grandparent = usr.
```

Figure 1: Result for `location(fstab, Parent)` query.

3 Question 3

Query the knowledge base to find all of the items directly inside the `/etc` folder. Ensure that you show the exact Prolog query and the results received

Result:

```
swipl
~/Flinders ML/AI/Lab08_Allia0024/Source git:(master) 18 files changed, 67 insertions(+), 1318 deletions(-)
swipl
?- location(What, root).
What = bin ;
What = etc ;
What =sbin ;
What = usr ;
What = var ;
What = home ;
What = lib ;
What = tmp .

?- file(crontab).
true.

?- folder(local).
true.

?- location(fstab, Parent).
ERROR: Syntax error: Operator expected
ERROR: location
ERROR: ** here **
ERROR: (fstab, Parent) .
?- location(fstab, Parent).
Parent = etc.

?- location(What, etc).
What = crontab ;
What = cups ;
What = fonts ;
What = fstab ;
What = 'host.conf' ;
What = bash.

?-
```

Figure 2: Result for `location(Item, etc)` query.

4 Question 4

First, define a new Prolog rule `is_in_grandparent (Item, GrandparentFolder)` that is true if an `Item` is located two levels deep inside the `GrandparentFolder`. Next, use this rule to find the grandparent folder of `/usr/local/bin`. Ensure to show the complete rule, the query, and the results.

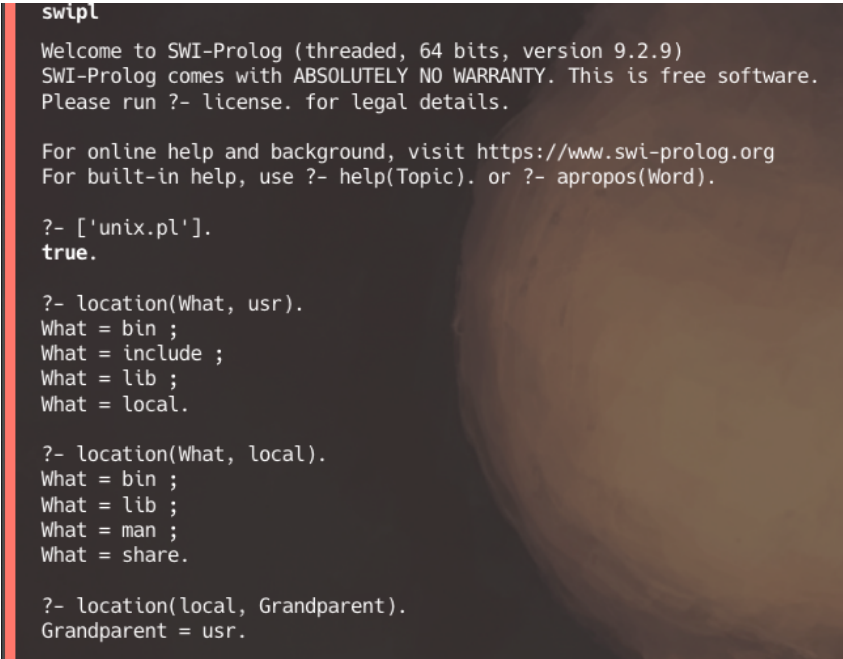
4.1 Implementation

```

1
2 is_in_grandparent(Item, GrandparentFolder) :-
3     location(Item, Parent),
4     location(Parent, GrandparentFolder).
5
6 find_path(Item, Path) :-
7     location(Item, Parent),
8     Parent \= root,
9     find_path(Parent, ParentPath),
10    append(ParentPath, [Parent], Path).

```

Result:



```

swipl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- ['unix.pl'].
true.

?- location(What, usr).
What = bin ;
What = include ;
What = lib ;
What = local.

?- location(What, local).
What = bin ;
What = lib ;
What = man ;
What = share.

?- location(local, Grandparent).
Grandparent = usr.

```

Figure 3: Result for `is_in_grandparent(bin, Grandparent)` query.

5 Question 5

Define a recursive Prolog rule `find_path(Item, Path)` that constructs the full, absolute path for any given `Item` as a list. The path should start with `root`. For example, querying for `crontab` should return `Path = [root, etc]`. Use your rule to find the path for the `/var/log` folder. Ensure to show the complete rule(s), the query, and the results.

5.1 Implementation

```

1 find_path(Item, [root]) :-                % Base case: path is [root]
2     location(Item, root).
3
4
5 find_path(Item, Path) :-
6     location(Item, Parent),
7     Parent \= root,
8     find_path(Parent, ParentPath),
9     append(ParentPath, [Parent], Path). % Append [Parent] to build path

```

Result:



```

~/Flinders ML/AI/Lab08_Alia0024/Source git:(master) 19 files changed, 1
swipl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
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Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- ['unix.pl'].
true.

?- find_path(bin, Path).
Path = [root] ;
Path = [root, usr] ;
Path = [root, usr, local] ;
false.

```

Figure 4: Result for `find_path(log, Path)` query.